

18. Perform Edge detection using Sobel Matrix along Y axis.

AIM:

Edge detection using Sobel Matrix along Y axis.

PROGRAM:

```
import cv2

img = cv2.imread(r'C:\Users\Susmitha\Downloads\ITA0510 Lab\expl.png', 0)

if img is None:
    print("Error: Image not found!") else:
    # Apply Sobel filter in the Y direction    sobel_y =
cv2.Sobel(img, cv2.CV_8U, 0, 1, ksize=5)

    # Save the output image    cv2.imwrite(
r'C:\Users\Susmitha\Downloads\ITA0510 Lab\sobel_y.jpg',
sobel_y
    )

    # Display the output
cv2.imshow("Original Image", img)
cv2.imshow("Sobel Y", sobel_y)

cv2.waitKey(0)
cv2.destroyAllWindows() INPUT:
```



OUTPUT:



RESULT:

Thus the program is executed successfully.